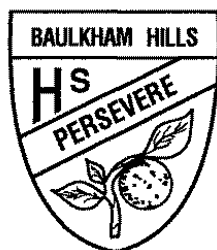


BAULKHAM HILLS HIGH SCHOOL



YEAR 12 TRIAL HSC EXAMINATION

2004

MATHEMATICS EXTENSION 1

*Time Allowed : Two hours
(Plus five minutes reading time)*

QUESTION 1

- (a) Find the co-ords of the point P that divides the interval A(-3, 4) and B(2, -3) externally in the ratio 1 : 2.

(b) Solve $\frac{4}{x-3} < 1$.

(c) Evaluate $\lim_{x \rightarrow 0} \frac{\sin 2x \cos 2x}{3x}$.

- (d) A curve has parametric equations $x = 2t - 2$, $y = t^2 + 1$. Find the cartesian equation for this curve.

(e) Use the substitution $u = 2 + x$ to evaluate $\int_2^2 x \sqrt{2+x} \, dx$.

QUESTION 2

(a) Find (i) $\int \tan x \, dx$.

(ii) $\int_{-3/2}^{3/4} \frac{1 \, dx}{\sqrt{9 - 4x^2}}$.

(b) Find the term independent of x in the binomial expansion $\left(x^2 + \frac{1}{x}\right)^9$.

(c) (i) Express $\sin 4t + \sqrt{3} \cos 4t$ in the form $R \sin(4t + \alpha)$, where α is in radians.

(ii) Hence solve $\sin 4t + \sqrt{3} \cos 4t = 0$ for $0 \leq t \leq \pi$.

QUESTION 3

(a)

Prove by induction $9^{n+2} - 4^n$ is divisible by 5 for $n \geq 1$.

(b) Consider the function $f(x) = 2 \tan^{-1}x$.

(i) State the range of the function $y = f(x)$.

(ii) Sketch the graph of $y = f(x)$.

(iii) Find the gradient of the tangent to the curve $y = f(x)$ at $x = \frac{1}{\sqrt{3}}$.

(c) (i) By equating the coefficients of $\sin x$ and $\cos x$, or otherwise, find constants A and B satisfying the identity.

$$A(2 \sin x + \cos x) + B(2 \cos x - \sin x) \equiv \sin x + 8 \cos x.$$

(ii) Hence evaluate $\int \frac{\sin x + 8 \cos x}{2 \sin x + \cos x} dx$.

QUESTION 4

(a)

If $x^3 - 8x^2 + kx - 12 = 0$ has one root equal to the sum of the other two; find k .

(b)

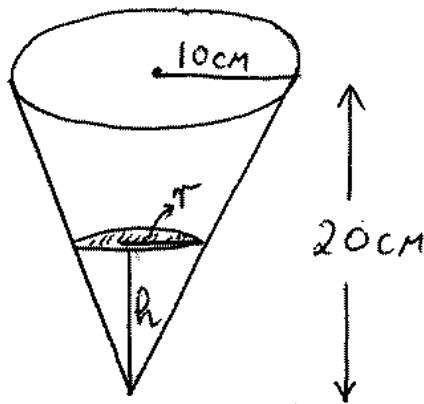
Taking $x = 0.5$ as the first approximation, use Newton's method to find a second approximation to the root of:

$$x - 3 + e^{2x} = 0.$$

Write your answer to 2 significant figures.

QUESTION 4 (Continued)

(c)



Water is poured into a conical vessel at a rate of $30\text{cm}^3/\text{s}$.

- (i) What is the rate of increase of the radius of water when $r = 5$.
- (ii) Hence find the rate of increase of the area of the surface of the liquid when $r = 5$.

(d) Using $\sin 3\theta = \sin (2\theta + \theta)$. Prove $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$.

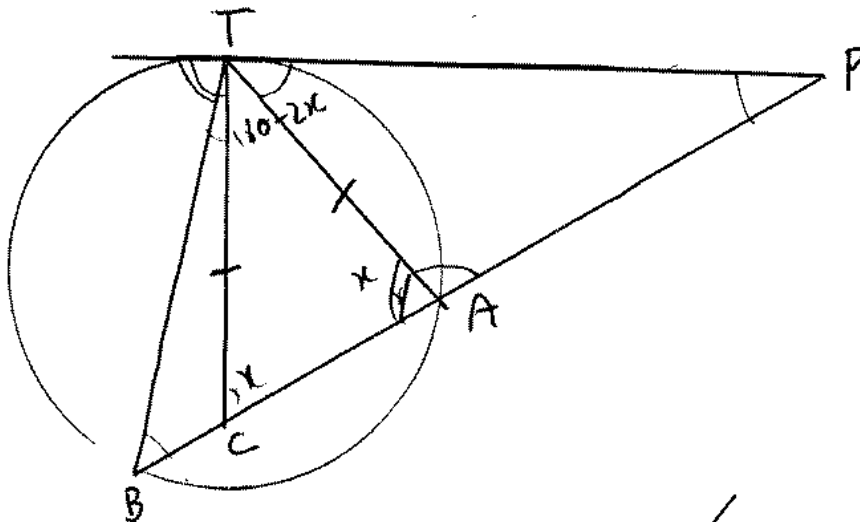
QUESTION 5

(a) A particle moves in a straight line such that its position x from a fixed point O at time ' t ' is given by $x = 5 + 8 \sin 2t + 6 \cos 2t$.

- (i) Prove the motion is simple harmonic motion.
- (ii) Find the period and amplitude of the motion.
- (iii) Find the greatest speed of the particle.

(b) State the largest positive domain for which $y = x^2 - 4x + 7$ has an inverse function.

(c)



PT is a tangent and PAB is a secant. $TC = TA$. Prove $\angle BTC = \angle TPA$

(d) By using the expansion $(1 + x)^n$. Prove $\sum_{k=0}^n 2^{3k} \binom{n}{k} = 3^{2n}$.

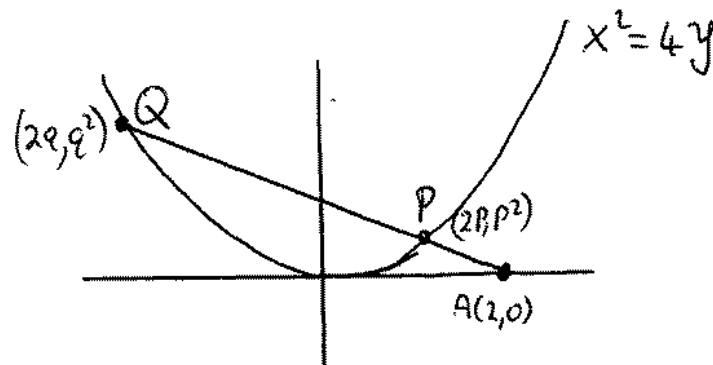
QUESTION 6

- (a) A particle is projected horizontally with velocity $V \text{ ms}^{-1}$, from a point h metres above the ground. Take $g \text{ ms}^{-2}$ as the acceleration due to gravity.
- (i) Taking the origin as the point on the ground immediately below the projection point, find expressions for x and y , the horizontal and vertical displacements of the particle at time ' t ' secs.
- (ii) Show the equation of the path is given by $y = \frac{2hV^2 - gx^2}{2V^2}$.
- (iii) Find the range of the particle.
- (b) Assume that the rate at which a body warms in air is proportional to the difference between its temperature T and the constant temperature A of the surrounding air. The rate can be expressed as:

$$\frac{dT}{dt} = K(T - A) \text{ where 't' is in minutes and K is constant.}$$

- (i) Show $T = A + Ce^{kt}$ (where C is constant) is a solution of the differential equation.
- (ii) A cooled body warms from 5°C to 10°C in 20 minutes. The air temperature is 20°C . Find the temperature of the body after a further 30 minutes have elapsed.
- (iii) Explain the behaviour of T as t becomes large.
- (c) Differentiate from 1st Principles $f(x) = x^2 - 2x + 1$.

QUESTION 7



- (a) The chord PQ joining the points $P(2p, p^2)$ and $Q(2q, q^2)$ on $x^2 = 4y$ always passes through the point $A(2, 0)$ when produced.
- (i) Show $(p + q) = pq$.
- (ii) Find the co-ordinates of M , the mid-point of PQ .
- (iii) Find the equation of the locus of M as P and Q vary on the parabola.

QUESTION 7 (Continued)

- (b) Two circles C_1 and C_2 are members of a set of circles defined by the equation:
 $x^2 + y^2 - 6x + 2ky + 3k = 0$ where k is real. The centre of C_1 lies on the line $x - 3y = 0$ and C_2 touches the x - axis. Find the equations of C_1 and C_2 .
- (c) Use Simpson's Rule with 3 function values to approximate the volume when $y = \ln x$ is rotated about the x - axis between $x = 1$ and $x = 3$.

YEAR 12 TRIAL EXT 1

MARKING SCALE

QUESTION 1

$$x = \frac{1 \times 2 + -2 \times -3}{1 - 2}$$

$$= -8$$

$$y = \frac{1 \times -3 + 4 \times -2}{-1 - 2}$$

$$= 11$$

$$\frac{4}{x-3} < 1$$

$$4(x-3) < (x-3)^2$$

$$0 < (x-3)(x-3-4)$$

$$0 < (x-3)(x-7)$$

$$\leftarrow \quad \quad \rightarrow$$

$$x < 3, \quad x > 7$$

$$\lim_{x \rightarrow 0} \frac{\sin 2x \cos 2x}{\sin 3x}$$

$$\lim_{x \rightarrow 0} \frac{\frac{1}{2} \sin 4x}{\sin 3x}$$

$$= \frac{2}{3}$$

$$x = 2t - 2, \quad y = t^2 + 1$$

$$t = \frac{x+2}{2}$$

$$y = \left(\frac{x+2}{2}\right)^2 + 1$$

$$e) \int_{-2}^2 x \sqrt{2+x} dx \quad u = 2+x$$

$$\frac{du}{dx} = 1$$

$$= \int_0^4 (u-2) \sqrt{u} du$$

$$= \int_0^4 u^{\frac{3}{2}} - 2u^{\frac{1}{2}} du$$

$$= \left[\frac{2}{5} u^{\frac{5}{2}} - \frac{4}{3} u^{\frac{3}{2}} \right]_0^4$$

$$= \left(\frac{2}{5} \cdot 32 - \frac{4}{3} \cdot 8 \right) - (0)$$

$$= 2\frac{2}{15}$$

Question 2.

$$1) \int \tan x dx = \int \frac{\sin x}{\cos x} dx$$

$$= -\ln(\cos x) + C$$

$$ii) \int_{-\frac{1}{2}}^{\frac{3}{4}} \frac{dx}{\sqrt{9-4x^2}} = \frac{1}{2} \int_{-\frac{1}{2}}^{\frac{3}{4}} \frac{dx}{\sqrt{\frac{9}{4}-x^2}}$$

$$= \frac{1}{2} \left[\sin^{-1} \frac{2x}{3} \right]_{-\frac{1}{2}}^{\frac{3}{4}}$$

$$= \frac{1}{2} \left[\sin^{-1} \left(\frac{1}{2} \right) - \sin^{-1} (-1) \right]$$

$$= \frac{1}{2} \left[\frac{\pi}{6} - -\frac{\pi}{2} \right]$$

$$= \frac{\pi}{3}$$

$$b) \left(x^2 + \frac{1}{x} \right)^9$$

$$T_{k+1} = {}^9C_k (x^2)^{9-k} \left(\frac{1}{x} \right)^k$$

$$= {}^9C_k x^{18-3k}$$

$$18-3k = 0$$

$$k = 6$$

$$T_7 = {}^9C_6$$

$$= 84$$

$$c) i) \sin 4t + \sqrt{3} \cos 4t = R \sin(4t + \alpha)$$

$$= R [\sin 4t \cos \alpha + \cos 4t \sin \alpha]$$

$$\therefore R \cos \alpha = 1$$

$$R \sin \alpha = \sqrt{3}$$

$$\therefore \tan \alpha = \sqrt{3}$$

$$\therefore \alpha = \frac{\pi}{3}$$

$$R = 2$$

$$\therefore \sin 4t + \sqrt{3} \cos 4t = 2 \sin(4t + \frac{\pi}{3})$$

$$ii) 2 \sin(4t + \frac{\pi}{3}) = 0$$

$$\therefore 4t + \frac{\pi}{3} = 0, \pi, 2\pi, 3\pi, 4\pi, \dots$$

$$4t = -\frac{\pi}{3}, \frac{2\pi}{3}, \frac{5\pi}{3}, \frac{8\pi}{3}, \frac{11\pi}{3}, \dots$$

$$t = -\frac{\pi}{12}, \frac{\pi}{6}, \frac{5\pi}{12}, \frac{2\pi}{3}, \frac{11\pi}{12}, \dots$$

$$\text{but } t \neq -\frac{\pi}{12}$$

4

12

Question 3

STEP 1: Prove True for $N=1$

$$9^3 - 4 = 725$$

which is divisible by 5.

$\therefore$ TRUE

Step 2: Assume True $N=K$

$$9^{K+2} - 4^K = 5A \text{ [A is INTEGER]}$$

Step 3: Prove True $N=K+1$

$$9^{K+2} - 4^{K+1} = 5B \text{ [B is INTEGER]}$$

$$\text{LHS} = 9^{K+2} - 4^{K+1}$$

$$= 9 \cdot 9^{K+1} - 4 \cdot 4^K$$

$$= 9(5A + 4^K) - 4 \cdot 4^K$$

$$= 45A + 5 \cdot 4^K$$

$$= 5(9A + 4^K)$$

Since A is integer, 4^K is integer

$$\therefore \text{LHS} = 5 \cdot B \text{ [B is integer]}$$

Step 4: If true $N=K$, true $N=K+1$

$\therefore$ Since True $N=1$, True $N=2$

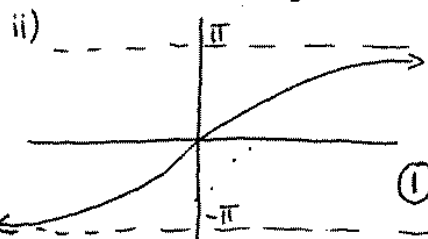
of True $N=2$, True $N=3$ etc

$\therefore$ True for all N . ①

4

b) i) $f(x) = 2 \tan^{-1} x$

Range: $-\pi \leq y \leq \pi$. ①



iii) $y = 2 \tan^{-1} x$

$$\frac{dy}{dx} = \frac{2}{1+x^2}$$

at $x = \frac{1}{\sqrt{3}}$

$$\frac{dy}{dx} = \frac{2}{1+\frac{1}{3}} = \frac{3}{2}$$

$\therefore$ Gradient of tangent = $\frac{3}{2}$. ① 4

c) $2A \sin x + A \cos x + 2B \cos x - B \sin x$

$$= \sin x + 8 \cos x$$

$$\begin{aligned} 2A - B &= 1 \\ A + 2B &= 8 \end{aligned}$$

$$\therefore \begin{aligned} A &= 2 \\ B &= 3 \end{aligned}$$

$$\begin{aligned} \therefore 2(2 \sin x + \cos x) + 3(2 \cos x - \sin x) \\ = \sin x + 8 \cos x \end{aligned}$$

$$\therefore \int \frac{\sin x + 8 \cos x}{2 \sin x + \cos x} dx = \int 2 + 3 \left(\frac{2 \cos x - \sin x}{2 \sin x + \cos x} \right) dx$$

$$= 2x + 3 \ln(2 \sin x + \cos x) + C. \quad ①$$

4

Question 4:

a) $x^3 - 8x^2 + Kx - 12 = 0$

Let roots be α, β, γ
where $\alpha = \beta + \gamma$.

$$\alpha + \beta + \gamma = 8$$

$$\therefore 2\alpha = 8$$

$$\alpha = 4.$$

$$\therefore 64 - 8(16) + 4K - 12 = 0$$

$$K = 19.$$

b) $Z_2 = Z_1 - \frac{f(Z_1)}{f'(Z_1)}$

$$f(x) = x - 3 + e^{2x}$$

$$f(0.5) = 0.218$$

$$f'(x) = 1 + 2e^{2x}$$

$$f'(0.5) = 6.437$$

$$Z_2 = 0.5 - \frac{0.218}{6.437}$$

$$= 0.47.$$

3

c) $\frac{dv}{dt} = 30$

$$V = \frac{1}{3} \pi r^2 h$$

$$\frac{r}{10} \therefore = \frac{h}{20}$$

$$\therefore h = 2r$$

$$\begin{aligned} V &= \frac{1}{3} \pi r^2 \cdot 2r \\ &= \frac{2}{3} \pi r^3 \end{aligned}$$

$$\frac{dv}{dt} = \frac{dv}{dr} \cdot \frac{dr}{dt}$$

$$30 = 2\pi r^2 \cdot \frac{dr}{dt}$$

$$r = 5$$

$$30 = 50\pi \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{3}{5\pi}$$

$$\frac{dA}{dt} = \frac{dA}{dr} \cdot \frac{dr}{dt}$$

$$A = \pi r^2$$

$$\frac{dA}{dt} = 2\pi r \cdot \frac{3}{5\pi} \quad (1)$$

$$= 10\pi \cdot \frac{3}{5\pi}$$

$$= 6 \text{ cm}^2/\text{s} \quad (5)$$

$$\sin 3\theta = \sin(2\theta + \theta)$$

$$= \sin 2\theta \cos \theta + \cos 2\theta \sin \theta$$

$$= 2\sin \theta \cos \theta \cdot \cos \theta \quad (1)$$

$$+ (1 - 2\sin^2 \theta) \sin \theta$$

$$\sin 3\theta = 2\sin \theta (1 - \sin^2 \theta) + \sin \theta - 2\sin^3 \theta$$

$$= 3\sin \theta - 4\sin^3 \theta \quad (2)$$

Question 5:

$$x = 5 + 8\sin 2t + 6\cos 2t$$

$$\dot{x} = 16\cos 2t - 12\sin 2t$$

$$\ddot{x} = -32\sin 2t - 24\cos 2t$$

$$= -4(8\sin 2t + 6\cos 2t) \quad (1)$$

$$= -4(x - 5) \quad (1)$$

since $x = -N(x - D)$ (5d)
 $\therefore$ SHM

$$\text{ii) Period} = \frac{2\pi}{\omega} = \pi \quad (1)$$

$$x = 5 + 8\sin 2t + 6\cos 2t$$

$$\text{but } 8\sin 2t + 6\cos 2t = 10\sin(2t + \phi) \quad (1)$$

$$\therefore \text{AMP} = 10 \quad (1)$$

$$\text{iii) } \dot{x} = 16\cos 2t - 12\sin 2t$$

$$= 20\cos(2t + \phi) \quad (1)$$

$$\therefore \text{greatest speed} = 20 \text{ m/s} \quad (6)$$

$$\text{b) } x \geq 2 \quad (1)$$

$$\angle TCA = \angle TAC = \gamma^\circ \text{ (base } \angle \text{ of } \triangle \text{ ISOS } \triangle)$$

$$\angle PTA = \angle TBC = \alpha^\circ$$

$$(L \text{ between tang. \& CHORD} = L \text{ in ALT. segment).} \quad (1)$$

$$\therefore \angle BTC = \gamma - \alpha \text{ (Ext } \angle \text{ of } \triangle = \text{sum of interior opp } \angle \text{ s)}$$

$$\angle TAP = (180 - \gamma) \text{ (L sum of st line)} \quad (1)$$

$$\therefore \angle TPA = \gamma - \alpha \text{ (L sum of } \triangle = 180^\circ)$$

$$\therefore \angle TPA = \angle BTC \quad (3)$$

$$(1+x)^N = {}^N C_0 + {}^N C_1 x + {}^N C_2 x^2 + \dots + {}^N C_N x^N$$

RTP:

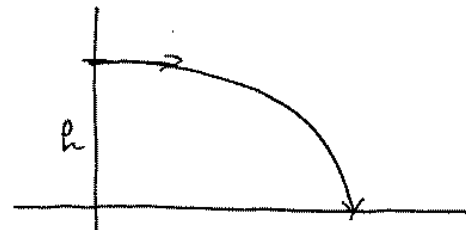
$${}^N C_0 + 8 {}^N C_1 + 64 {}^N C_2 + \dots + 2^{3N} {}^N C_N = 3^{2N} \quad (1)$$

$$\text{Let } x = 8$$

$$9^N = {}^N C_0 + 8 {}^N C_1 + 64 {}^N C_2 + \dots \quad (1)$$

$$\therefore 3^{2N} = {}^N C_0 + 8 {}^N C_1 + 64 {}^N C_2 + \dots \quad (2)$$

Question 6:



$$\ddot{x} = 0$$

$$\ddot{y} = -g$$

$$\dot{x} = V \cos \alpha$$

$$\dot{y} = -gt + V \sin \alpha$$

$$\alpha = 0$$

$$\alpha = 0$$

$$\dot{x} = V$$

$$\dot{y} = -gt$$

$$x = Vt + C$$

$$t = 0, x = 0 \therefore C = 0 \quad y = -gt^2/2 + C$$

$$x = Vt$$

$$t = 0, y = h$$

$$\therefore C = h$$

$$y = -gt^2/2 + h \quad (1)$$

$$\text{iv } t = \frac{1}{V} \quad (1)$$

$$y = -g \left(\frac{1}{V} \right)^2 + h$$

$$= -\frac{g}{2V^2} + h$$

$$= \frac{2V^2 h - gx^2}{2V^2} \quad (1)$$

$$\text{iii) Range: } x \text{ when } y = 0$$

$$y = -\frac{gt^2}{2} + h$$

$$0 = -\frac{gt^2}{2} + h$$

$$t = \sqrt{\frac{2h}{g}} \quad (1)$$

$$\therefore x = V \sqrt{\frac{2h}{g}} \quad (1)$$

(6)

$$\text{b) } T = A + Ce^{kt}$$

$$\frac{dT}{dt} = kC e^{kt}$$

$$= k(T - A)$$

$$\text{but } Ce^{kt} = T - A$$

$$\frac{dT}{dt} = k(T - A) \quad (1)$$

$$T = 20 + Ce^{kt}$$

$$t=0 \quad T=5$$

$$5 = 20 + Ce^0$$

$$\therefore C = -15 \quad (1)$$

$$T = 20 - 15e^{kt}$$

$$t = 20, \quad T = 10$$

$$10 = 20 - 15e^{20k}$$

$$15e^{20k} = 10$$

$$20k = \ln\left(\frac{2}{3}\right)$$

$$k = -0.02 \quad (1)$$

$$T = 20 - 15e^{-0.02 \times 50}$$

$$= 14.48^\circ \quad (1)$$

$$\text{iii) } t \rightarrow \infty, \quad T \rightarrow 20^\circ \quad (1)$$

$$9) \quad f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(x+h)^2 - 2(x+h) - 1 - (x^2 - 2x + 1)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2xh + h^2 - 2h}{h} \quad (1)$$

$$= \lim_{h \rightarrow 0} 2x + h - 2 \quad (1)$$

$$= 2x - 2$$

(B)

(2)

Question 7.

$$M_{PQ} = \frac{p^2 - q^2}{2p - 2q}$$

$$= \frac{(p-q)(p+q)}{2(p-q)}$$

$$= \frac{p+q}{2} \quad (1)$$

EQN CHORD PQ

$$y - p^2 = \frac{p+q}{2} (x - 2p) \quad (1)$$

Since (2, 0) lies on line

$$-p^2 = \frac{p+q}{2} (2 - 2p)$$

$$-2p^2 = 2p + 2q - 2p^2 - 2pq$$

$$2pq = 2(p+q)$$

$$\therefore pq = p+q \quad (1)$$

$$\text{ii) } \left. \begin{aligned} x &= p+q \\ y &= \frac{p^2+q^2}{2} \end{aligned} \right\} \quad (1)$$

$$\text{iii) } p^2 + q^2 = (p+q)^2 - 2pq$$

$$2y = x^2 - 2x \quad (1)$$

(5)

7b

$$x^2 + y^2 - 6x + 2ky + 3k = 0.$$

$$x^2 - 6x + 9 + y^2 + 2ky = -3k + 9.$$

CIRCLE 1:

x co-ord centre: 3. (1)

Since lies on $x = 3y$

$$\therefore y = 1$$

$\therefore$ centre (3, 1).

$$\text{CIRCLE 1: AS: } (x-3)^2 + (y-1)^2 = r^2$$

$$\therefore +2ky = -2y$$

$$\therefore k = -1 \quad (1)$$

$$\text{EQN CIRCLE: } x^2 + y^2 - 6x + 2y + 3 = 0.$$

CIRCLE 2: (3, 0) lies on the circle.

$$\therefore 9 - 18 + 3k = 0$$

$$k = 3 \quad (1)$$

EQN CIRCLE:

$$x^2 + y^2 - 6x + 6y + 9 = 0.$$

$$(4) \quad (1)$$

$$V = \pi \int y^2 dx.$$

$$= \pi \int_1^3 (\ln x)^2 dx \quad (1)$$

$\frac{1}{2} (\ln x)^2$			
x	1	2	3
y	0	0.48	1.21

$$V = \pi \cdot \frac{1}{3} [0 + 4 \times 0.48 + 1]$$

$$= 1.04 \pi$$

$$= 3.28 \quad (1)$$

(3)

(12)